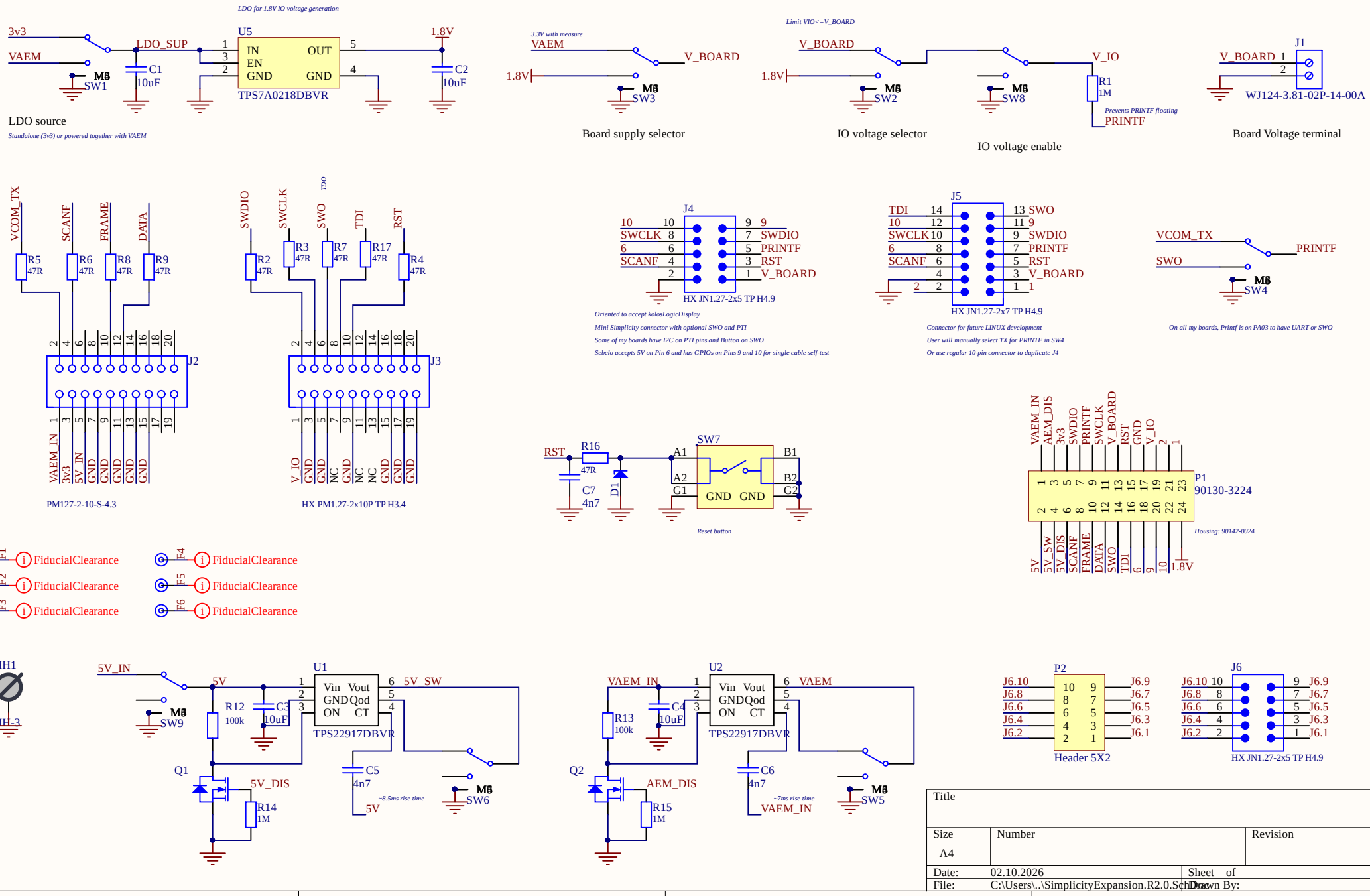
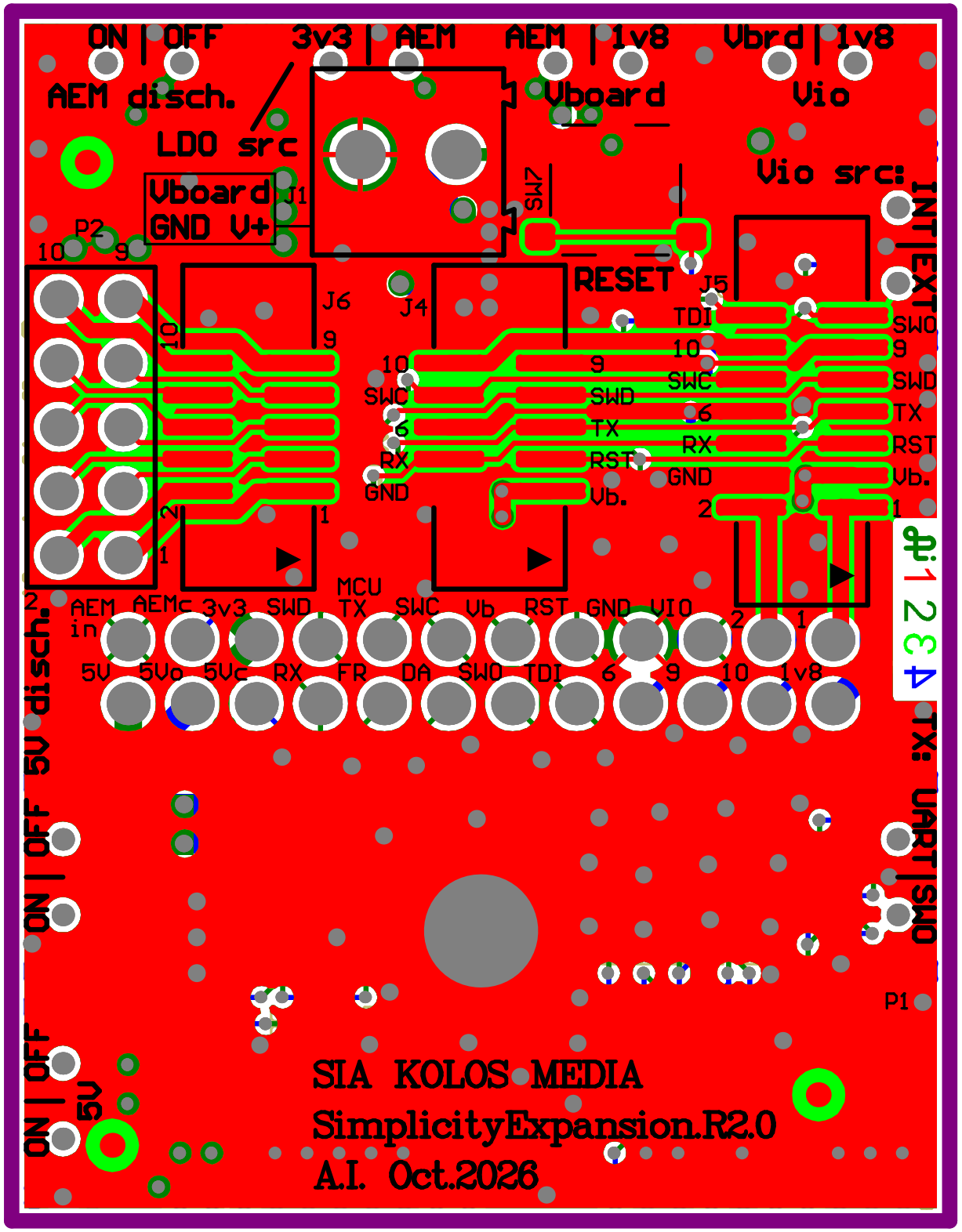
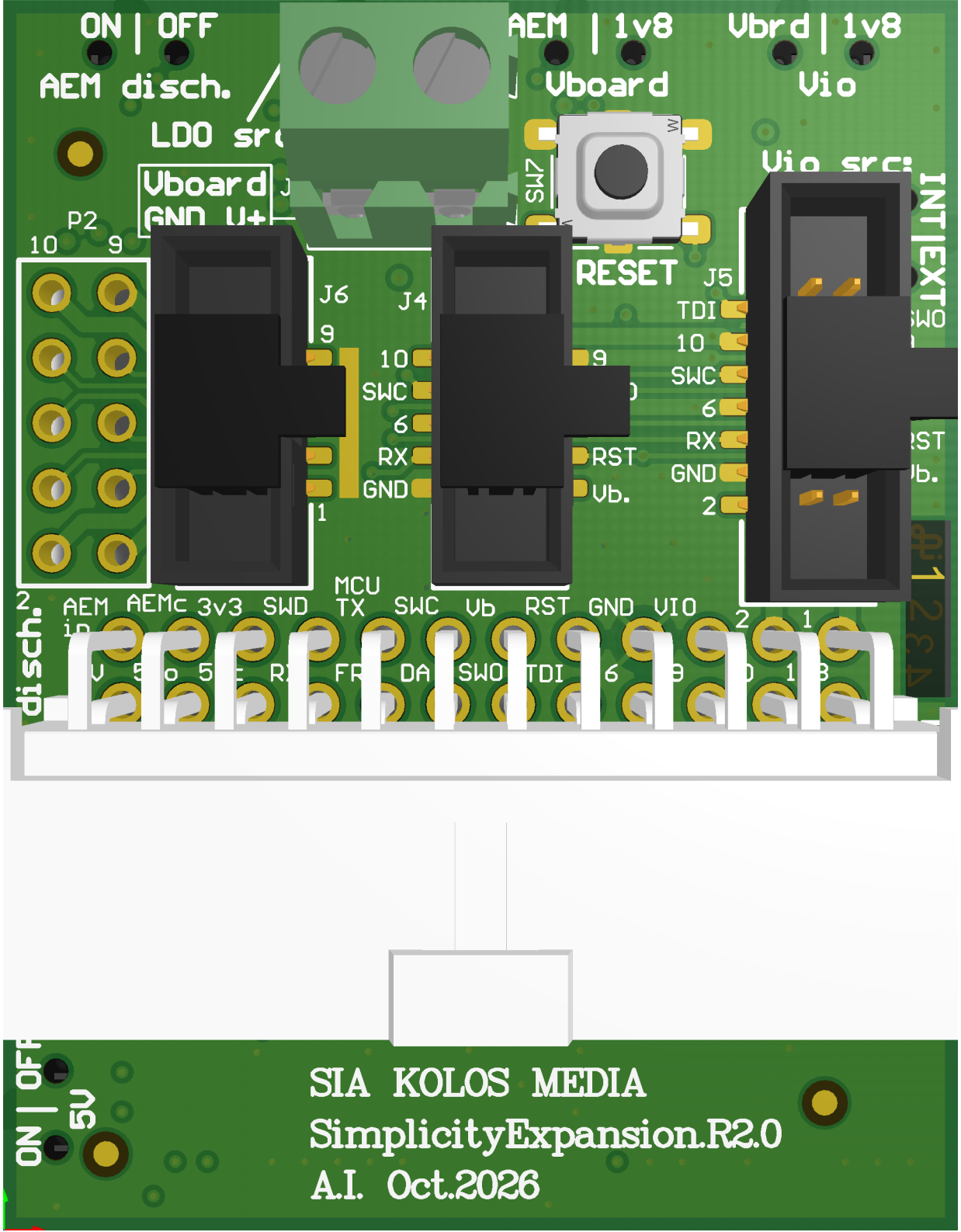
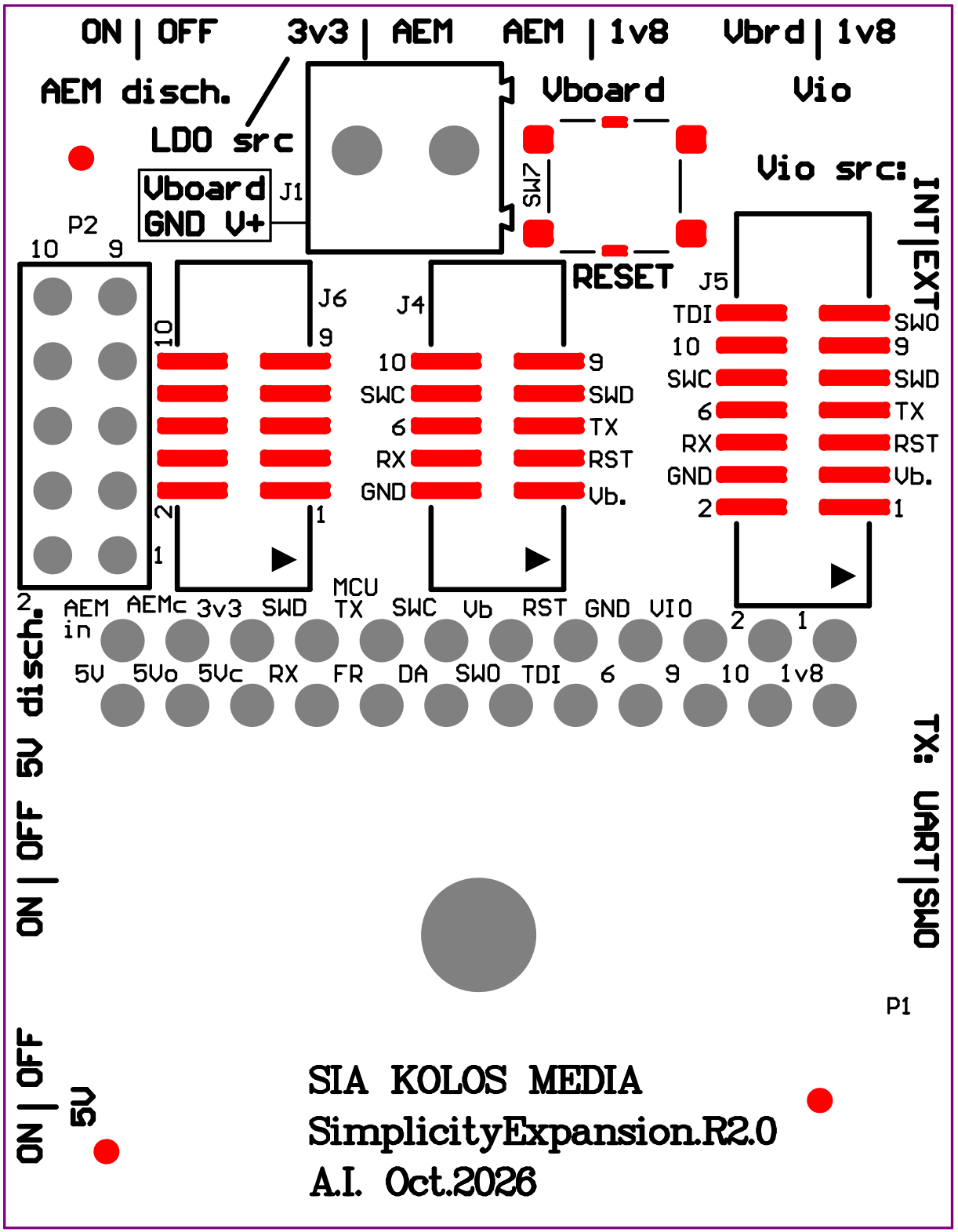
Designator	Comment	Description	Footprint	Manufacturer	Part Number	Price 1	Price 100	Price 1000	Quantity
C1, C2, C3, C4	CL10A106KP8NNNC	10uF, 6.3V, 0603, X5R, 20%	C SMD Ceramic 0603	Wurth Elektronik	885012106006	0.152	0.152	0.152	4
C5, C6, C7	0603B472K500NT	4.7nF, 50V, 0603, X7R, 10%	C SMD Ceramic 0603	Wurth Elektronik	885012206087	0.03	0.03	0.03	3
D1	H5VU10U	TVS diode, DFN1006-2, 0.6pF, 5V	DFN1006	BORN	ESD9L5.0ST5G-N	0.03	0.03	0.03	1
J1	WJ124-3.81-02P-14-00A	WR-TBL Serie 2143 Horizontal Entry Modular, Rising Cage Clamp, pitch 3.81mm, 2p	691214310002	Wurth Elektronik	691214310002	0.5	0.5	0.5	1
J2	PM127-2-10-S-4.3	Socket Header, SMT, pitch 1.27mm, Dual Row, Vertical, 20p	62332021021	Wurth Elektronik	62332021021	0.5	0.4	0.3	1
J3	HX PM1.27-2x10P TP H3.4	Socket Header, SMT, pitch 1.27mm, Dual Row, Vertical, 20p, H=3.4	62332021021	hanxia	HX PM1.27-2x10P TP H3.4	0.5	0.4	0.3	1
J4, J6	HX JN1.27-2x5 TP H4.9	WR-BHD Box Header, Male PCB side, pitch 1.27mm, SMT, Angled, 10p	62731020621	Wurth Elektronik	62731020621	0.3	0.3	0.3	2
J5	HX JN1.27-2x7 TP H4.9	WR-BHD Box Header, Male PCB side, pitch 1.27mm, SMT, Angled, 14p	62731420621	Wurth Elektronik	62731420621	0.5	0.4	0.3	1
P1	90130-3224	Header, 12-Pin, Dual row	Molex 24 pin 90130-3224	Molex	90130-3224	6	5	4	1
Q1, Q2	HL2312	N-MOSFET, SOT23, 20V, 5.5A, 1.8V operation	SOT23-3-IPC - 1mm	Nexperia	PMV16XN	0.08	0.07	0.06	2
R1, R14, R15	0603WAF1004T5E	1MOhm, 50 V, 0603, 100 mW, ±1%	R SMD 0603 1%	VISHAY	CRCW06031M00FKTABC	0.0123	0.0123	0.0026	3
R2, R3, R4, R5, R6, R7, R8, R9, R16, R17	0603WAF470JT5E	47R, 50 V, 0603, 100mW, ±1%	R SMD 0603 1%	PANASONIC	ERJ-3EKF47R0V	0.09	0.015	0.009	10
R12, R13	0603WAF1003T5E	100k, 50 V, 0603, 100mW, ±1%	R SMD 0603 1%	Royal Ohm	0603WAF1003T5E	0.09	0.015	0.009	2
SW1, SW2, SW3, SW4, SW5, SW6, SW8, SW9	MK-12C02-G020	Slide Switch, SPDT	450404015514	Wurth Elektronik	450404015514	0.13	0.1	0.08	8
SW7	TS-1187A-B-A-B	Tact Switch with ground terminal, 5x5mm, 160g	431181015816	Wurth Elektronik	431181015816	0.3	0.3	0.3	1
U1, U2	TPS22917DBVR	Load Switch, 1–5.5V, 2-A, 80-mO, Ultra-Low Leakage	SOT23-6BL	Texas Instruments	TPS22917DBVR	0.5	0.35	0.17	2
U5	TPS7A0218DBVR	200mA, nanopower-IQ (25 nA), low-dropout (LDO) voltage regulator with enable	SOT23-5BL	Texas Instruments	TPS7A0218DBVR	0.915	0.64	0.417	1


LDO for 1.8V IO voltage generation
3v3
VAEM
LDO_SUP
C1
10uF
SW1
U5
IN
EN
GND
OUT
GND
TPS7A0218DBVR
1.8V
C2
10uF
LDO source
Standalone (3v3) or powered together with VAEM
3.3V with measure
VAEM
V_BOARD
1.8V
SW3
Board supply selector
Limit VIO<=V_BOARD
V_BOARD
V_IO
R1
1M
SW2
SW8
Prevents PRINTF floating
PRINTF
IO voltage selector
IO voltage enable
J1
V_BOARD
WJ124-3.81-02P-14-00A
Board Voltage terminal
VCOM_TX
SCANF
FRAME
DATA
R5 47R
R6 47R
R8 47R
R9 47R
J2
VAEM_IN
3v3
5V_IN
PM127-2-10-S-4.3
SWDIO
SWCLK
SWO
TDI
RST
R2 47R
R3 47R
R7 47R
R17 47R
R4 47R
J3
V_IO
HX PM1.27-2x10P TP H3.4
J4
SWCLK
SWDIO
PRINTF
RST
V_BOARD
HX JN1.27-2x5 TP H4.9
Oriented to accept kolosLogicDisplay
Mini Simplicity connector with optional SWO and PTI
Some of my boards have I2C on PTI pins and Button on SWO
Sebelo accepts 5V on Pin 6 and has GPIOs on Pins 9 and 10 for single cable self-test
J5
TDI
SWO
SWCLK
SWDIO
PRINTF
RST
V_BOARD
HX JN1.27-2x7 TP H4.9
Connector for future LINUX development
User will manually select TX for PRINTF in SW4
Or use regular 10-pin connector to duplicate J4
VCOM_TX
SWO
PRINTF
SW4
On all my boards, Printf is on PA03 to have UART or SWO
RST
R16
47R
C7
4n7
D1
SW7
A1
A2
G1
B1
B2
G2
GND GND
Reset button
VAEM_IN
AEM_DIS
3v3
SWDIO
PRINTF
SWCLK
V_BOARD
RST
GND
V_IO
P1
90130-3224
Housing: 90142-0024
5V
5V_SW
5V_DIS
SCANF
FRAME
DATA
SWO
TDI
1.8V
F1 FiducialClearance
F2 FiducialClearance
F3 FiducialClearance
F4 FiducialClearance
F5 FiducialClearance
F6 FiducialClearance
MH1
MH-3
5V_IN
U1
Vin Vout
GND Qod
ON CT
TPS22917DBVR
5V_SW
R12
100k
C3
10uF
SW9
Q1
5V_DIS
R14
1M
C5
4n7
~8.5ms rise time
SW6
VAEM_IN
U2
VAEM
C4
10uF
R13
100k
Q2
AEM_DIS
R15
1M
C6
4n7
~7ms rise time
SW5
P2
J6.10
J6.8
J6.6
J6.4
J6.2
J6.9
J6.7
J6.5
J6.3
J6.1
Header 5X2
J6
HX JN1.27-2x5 TP H4.9
Title
Size
A4
Number
Revision
Date:
02.10.2026
Sheet of
File:
C:\Users\..\SimplicityExpansion.R2.0.SchDoc
Drawn By:


ON | OFF
3v3 | AEM
AEM | 1v8
Vbrd | 1v8
AEM disch.
LDO src
Vboard
Vio
Vio src:
Vboard
GND V+
J1
SW7
P2
10
9
J6
J4
RESET
J5
TDI
SWO
SWC
SWD
TX
RX
RST
GND
Vb.
MCU
AEM in
AEMc
3v3
SWD
TX
SWC
Vb
RST
GND
VIO
5V
5Vo
5Vc
RX
FR
DA
SWO
TDI
6
9
10
1v8
INT|EXT
UART|SWO
ON | OFF 5V disch.
ON | OFF
5V
P1
SIA KOLOS MEDIA
SimplicityExpansion.R2.0
A.I. Oct.2026


ON | OFF
AEM disch.
LDO src
Vboard
GND V+
P2
10
9
2
1
J6
J4
J5
9
10
SWC
6
RX
GND
RST
Vb.
TDI
10
SWC
6
RX
GND
2
RESET
SW7
AEM | 1v8
Vboard
Vbrd | 1v8
Vio
Vio src:
INT|EXT
MCU
AEM disch.
AEMc
3v3
SWD
TX
SWC
Vb
RST
GND
VIO
ON | OFF
5V
SIA KOLOS MEDIA
SimplicityExpansion.R2.0
A.I. Oct.2026


ON | OFF
3v3 | AEM
AEM | 1v8
Vbrd | 1v8
AEM disch.
LDO src
Vboard
Vio
Vboard
GND V+
J1
SW7
Vio src:
INT | EXT
P2
10
9
J6
J4
RESET
J5
TDI
SWO
10
9
SWC
SWD
6
TX
RX
RST
GND
Vb.
2
1
MCU
AEM in
AEMc
3v3
SWD
TX
SWC
Vb
RST
GND
VIO
5V
5Vo
5Vc
RX
FR
DA
SWO
TDI
6
9
10
1v8
ON | OFF 5V disch.
TX: UART | SWO
P1
ON | OFF
5V
SIA KOLOS MEDIA
SimplicityExpansion.R2.0
A.I. Oct.2026

GND
P2=16
NO=0 LTC
FFO=1
1 2 3 4 5 6 7 8 9 10
NC I AEM_MEM AEM_LTC 3v3 SMDIO PRINTF=TXT SMCLK V_BOARD RESET GND V_IO 5 1
IDT
ATAD
EMART
XA=ANCS
LTC_VS
fun_VS
VS
8v1
5
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16